Ogbonna Arinze

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SUMMARY

Fullstack JavaScript Engineer with 8 years of professional experience building and shipping web applications from the ground up across ad-tech-adjacent industries including SaaS platforms, fintech, healthcare, and civic tech. Expert-level proficiency in React, Node.js, Next.js, and TypeScript, with production experience across the full stack: frontend architecture, backend APIs, database design (MongoDB, PostgreSQL), cloud infrastructure (AWS, Docker), and AI-powered product features. Proven track record of owning entire product builds end-to-end, making architectural decisions independently, and thriving in small, fast-moving teams with startup-level ownership.

CORE COMPETENCIES

- **Fullstack JavaScript:** React, Next.js, Node.js, NestJS, TypeScript, modern JavaScript (ESNext), Redux, Zustand, and Vue.js.
- **Backend & APIs:** NestJS, Node.js REST APIs, JWT authentication, role-based access control (RBAC), GraphQL, and server-side rendering (SSR).
- **Databases:** MongoDB (schema design, data modeling), PostgreSQL (pgvector, Prisma ORM), and MySQL familiarity.
- **Cloud & Infrastructure:** AWS (CloudFront, S3), Docker/Docker Compose, GitHub Actions CI/CD, Heroku, Linux, Nx monorepo orchestration, and Traefik reverse proxy.
- **AI & Automation:** RAG pipelines, OpenAI API, LangChain, Vercel AI SDK, Mastra multi-agent framework, vector search, and embeddings.
- **Frontend Architecture:** Scalable component systems, design systems (Radix UI), Tailwind CSS, Storybook, performance optimization, and Webpack/Vite/Rollup.
- **Testing & Quality:** Playwright, Jest, Cypress, Allure Reports, ESLint, Prettier, and CI-driven QA.
- **Leadership:** Technical architecture decisions, mentorship, Agile/Scrum, and product-minded development in small teams.

WORK EXPERIENCE

Genentech (Roche) San Francisco, CA (Remote)
Frontend Engineer *May 2023 - Present*

- Spearheaded the redesign of the main clinical trial data product, migrating from a **Vue.js** web application to a **React Chrome extension**, making key architecture decisions on component structure, state management, and build tooling.
- Co-developed a full-stack **Next.js chat widget** powered by a **RAG system** using **OpenAI GPT models**, **LangChain**, **Vercel AI SDK**, and **Chroma** vector database, enabling users to connect data sources and query them in real time via a conversational interface.
- Implemented **JWT-based authentication** and **role-based access control** on the frontend using HTTP-only cookies for secure token handling, collaborating closely with backend engineers to align on auth strategies and permission-based UI rendering.
- Architected a **reusable design system** in **React** and **Radix UI** with **Tailwind CSS**, improving UI consistency, accessibility (WCAG), and developer velocity across multiple product teams.
- Introduced **Redux** for state management and **React Query** for request caching, refactoring legacy class components to functional components and optimizing the build pipeline.
- Optimized application performance by implementing **lazy loading, code splitting**, and **CDN asset delivery**, reducing load times significantly as measured via **Lighthouse** and Chrome DevTools.
- Increased release reliability by integrating **Playwright end-to-end tests** into the **CI/CD pipeline** with **GitHub Actions**, automating reporting with **Allure Reports** on GitHub Pages and configuring Slack notifications for build visibility.
- Mentored a team of two junior developers (interns), fostering a culture of technical excellence and code quality.

Descasio Lagos, Nigeria (Hybrid)
Senior Frontend Developer *May 2021 - Mar 2023*

- Led the design and implementation of an intuitive **React** user interface, achieving a **25% boost in user engagement** and collaborating with architects to create a scalable frontend architecture that accelerated development by **30%**.
- Optimized frontend performance by profiling render bottlenecks with **React Profiler** and **Lighthouse**, reducing bundle size by **30-40%** through rollup-plugin-visualizer and webpack-bundle-analyzer, achieving a **35% faster load time**.
- Drove a **40% improvement** in application performance by optimizing caching mechanisms across the React application.
- Enhanced development workflow by introducing **Jest** and **Enzyme** testing frameworks, leading to a **20% decrease** in bug reports.
- Established engineering standards using **ESLint, Prettier, Husky, Lint-staged**, and **GitHub Actions**, reducing errors and accelerating CI/CD processes.
- Mentored junior engineers on frontend best practices, fostering high-quality, maintainable code across projects.

ActivEdge Technologies Lagos, Nigeria
Senior React Developer *Oct 2019 - Sep 2021*

- Built a high-volume **bank Settlement Solution** in a team of 3 frontend engineers, handling thousands to millions of daily transactions and boosting settlement rates by **80%**.
- Developed critical payment modules (MasterCard, VISA, Express-cash, E-commerce) using **React** with **Redux** for state management, **RxJS** for data streaming, and **Axios** for asynchronous API requests.
- Built a real-time account reconciliation solution, boosting daily reconciliation by **60%** through innovative transaction processing and data management strategies.
- Adopted the native **EventSource API** for real-time data streaming, replacing socket.io for faster updates and a more responsive interface.
- Built dynamic data visualization dashboards using **React** and **react-chartjs**, streamlining management of user roles and permissions with **Redux Toolkit** and **Ant Design**.

Founders Lair Beijing, China (Remote)
Full Stack Engineer (Contract) *Oct 2020 - Feb 2021*

- Developed **Node.js RESTful APIs** and translated business requirements into efficient **MongoDB** database models and schemas, collaborating closely with the business team to ensure technical solutions met product needs.
- Transformed Figma designs into reusable **React** components, ensuring a consistent, intuitive user interface across the application.
- Engineered a **CI/CD pipeline** for automated deployment of APIs to testing environments on **Heroku** and production servers on **AWS**.
- Mentored two interns through frontend and backend projects, providing technical guidance and professional development support.

SureTrade Business Solutions Nigeria
Frontend Developer *Mar 2018 - Oct 2019*

- Designed **MongoDB** database schemas and implemented server-side logic in **Node.js**, building and testing **REST API** endpoints for multiple client-facing applications.
- Developed **SureEdu**, a school management portal in a team of 5, handling staff payroll and attendance tracking, student records (exams, reports, attendance), and guardian communications.
- Responsible for full-stack integration, connecting the **JavaScript** frontend to **Node.js** backend services, developing user use cases, and converting business requirements into database schemas and models.
- Devised comprehensive documentation for each application, detailing all functions, capabilities, and features. Tested user-defined functions, triggers, and API endpoints to ensure reliability.

PROJECTS

OurNigeria ournigeria.ng
Senior Engineer *Feb 2026 - Present*

AI-powered civic tech SaaS platform for Nigerian budget analysis. 700+ documents, 708K+ vector embeddings, 37 states.

- Architected and built the entire platform from scratch as an **Nx monorepo** managing **6 applications**: **NestJS** API backend, **Next.js** citizen-facing web app with 22 interactive chart types (Recharts), Next.js admin dashboard, NestJS document ingestion pipeline, Next.js marketing landing page, and a Remotion video generation service.
- Designed the **PostgreSQL 16** database schema with **Prisma v7**, including pgvector extension for vector storage, and built the **Node.js** backend API with **NestJS** handling authentication (phone OTP, Telegram OAuth), session management, and RESTful endpoints.
- Built a **multi-agent AI system** using **Mastra** with 5 specialized agents and a three-stage intent classification pipeline, powered by a hybrid RAG pipeline combining **pgvector** + BM25 full-text search with Reciprocal Rank Fusion and **Cohere** reranking.
- Implemented **SSE streaming** for real-time chat responses with keepalive heartbeats, token-budgeted context management, and conversation summarization for memory-efficient long conversations.
- Configured **CI/CD** with **GitHub Actions** for automated builds, **Playwright** E2E tests across 4 projects, and **Docker Compose** production deployment with multi-stage Dockerfiles, health checks, memory limits, and **Traefik** reverse proxy routing on **Linux** servers.

EDUCATION

University of Nigeria Nsukka, Enugu
BSc Computer Science - 4.1 CGPA *Graduation Date: Sep 2019*